

MAHALAKSHMI SABANAYAGAM

[Email](#) | [Google Scholar](#) | [Website](#) | [Github](#) | [LinkedIn](#)

RESEARCH INTEREST

I am interested in the theory of machine/deep learning, primarily in understanding its connection to kernels and the interplay between adversarial robustness and optimization. I also focus on graph based learning problems.

EDUCATION

Research Fellow AIML, University of Adelaide Advisor: <i>Prof. Simon Lucey</i>	February 2026 – January 2028
Ph.D. in Computer Science Technical University of Munich, Germany (magna cum laude) Thesis: Robust Machine Learning from the Theory of Overparameterization Advisor: <i>Prof. Debarghya Ghoshdastidar</i>	August 2021 – December 2025
Master of Science , Informatics Technical University of Munich, Germany	October 2018 – June 2021 <i>CGPA: 1.3 (best of 1.0)</i>
Bachelor of Technology , Computer Science & Engineering National Institute of Technology, Trichy, India	July 2011 – May 2015 <i>CGPA: 9.37 (best of 10)</i>

ACADEMIC / INDUSTRY EXPERIENCE

Research Visitor , Monash University, Australia Worked with <i>Dr. Russell Tsuchida</i> and <i>Dr. Dinh Phung</i> on theoretical analysis of adversarial robustness of probabilistic models.	August 2025 – November 2025
Research Visitor , CSIRO, Australia Worked with <i>Dr. Cheng Soon Ong</i> and <i>Dr. Russell Tsuchida</i> , on theoretical analysis of adversarial robustness of probabilistic models.	August 2024 – November 2024
Research Visitor , New York University, USA Worked with <i>Prof. Julia Kempe</i> , on theoretical analysis of adversarial robustness of neural networks, and together with <i>Prof. Andrew Gordon Wilson</i> on robustness to distributional shifts under Bayesian inference.	March 2023 – June 2023
Computer Scientist 1 , Adobe Systems, India Developed a robust OS agnostic (Mac/Windows) framework for Dreamweaver, with HiDPI adaptation. Upgraded Chromium Embedded Framework (CEF) with custom optimization for messaging queue and memory. Recognized as a top contributor and was awarded two early promotions - <i>Member of Technical Staff 2</i> in January 2017 and <i>Computer Scientist 1</i> in January 2018.	July 2015 – September 2018
Research Intern , Samsung R&D Institute, India Implemented a module for secure log-out in Android Browser of Samsung. Worked on improving the efficiency of Optical Character Recognition using Tesseract and OpenCV.	May 2014 – July 2014

PUBLICATIONS

12. **Robustness Certificates for Neural Networks against Data Poisoning and Evasion Attacks** [\[paper\]](#)
Sara Taheri, Mahalakshmi Sabanayagam, Debarghya Ghoshdastidar, Majid Zamani
IEEE Open Journal of Control Systems, Special Section: Intersection of Machine Learning with Control, 2026
11. **Exact Certification of (Graph) Neural Networks Against Label Flipping** [\[paper\]](#)[\[code\]](#)
Mahalakshmi Sabanayagam*, Lukas Gosch*, Stephan Günnemann, Debarghya Ghoshdastidar
Spotlight at International Conference on Learning Representations (*ICLR 2025*)
Oral at the VerifAI: AI Verification in the Wild Workshop, ICLR (*VerifAI ICLR 2025*)

*Authors contributed equally to the work

10. **Provable Robustness of (Graph) Neural Networks Against Data Poisoning and Backdoor Attacks** [\[paper\]](#)[\[code\]](#)
 Lukas Gosch*, *Mahalakshmi Sabanayagam**, Debarghya Ghoshdastidar, Stephan Günnemann
 Transactions on Machine Learning Research (*TMLR 2025*)
Oral, Best Paper Award at New Frontiers of Adversarial Machine Learning Workshop, NeurIPS (*AdvML-Frontiers NeurIPS 2024*)
9. **Kernels, Data & Physics** [\[paper\]](#)
 Francesco Cagnetta, Deborah Oliveira, *Mahalakshmi Sabanayagam*, Nikolaos Tsilivis, Julia Kempe
 Journal of Statistical Mechanics: Theory and Experiment (*JSTAT Lecture Notes 2024*)
8. **Robust Feature Inference: A Test-time Defense Strategy using Spectral Projections** [\[paper\]](#)[\[code\]](#)
 Anurag Singh*, *Mahalakshmi Sabanayagam**, Krikamol Muandet, Debarghya Ghoshdastidar
 Transactions on Machine Learning Research (*TMLR 2024*)
7. **Unveiling the Hessian’s Connection to the Decision Boundary** [\[paper\]](#)[\[code\]](#)
Mahalakshmi Sabanayagam, Freya Behrens, Urte Adomaityte, Anna Dawid
 Mathematics of Modern Machine Learning Workshop, NeurIPS (*M3L NeurIPS 2023*)
6. **Analysis of Convolutions, Non-linearity and Depth in Graph Neural Networks using Neural Tangent Kernel** [\[paper\]](#)[\[code\]](#)
Mahalakshmi Sabanayagam, Pascal Esser, Debarghya Ghoshdastidar
 Transactions on Machine Learning Research (*TMLR 2023*)
5. **Improved Representation Learning Through Tensorized Autoencoders** [\[paper\]](#)[\[code\]](#)
 Pascal Esser*, Satyaki Mukherjee*, *Mahalakshmi Sabanayagam**, Debarghya Ghoshdastidar
 International Conference on Artificial Intelligence and Statistics (*AISTATS 2023*)
4. **Analysis of Graph Convolution Networks using Neural Tangent Kernels** [\[paper\]](#)[\[code\]](#)
Mahalakshmi Sabanayagam, Pascal Esser, Debarghya Ghoshdastidar
 MLG workshop, European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (*ECML PKDD 2022*)
3. **Graphon based Clustering and Testing of Networks: Algorithms and Theory** [\[paper\]](#)[\[code\]](#)
Mahalakshmi Sabanayagam, Leena Chennuru Vankadara, Debarghya Ghoshdastidar
 International Conference on Learning Representations (*ICLR 2022*)
2. **Rough Set-based Feature Selection for Credit Risk Prediction using Weight Adjusted Boosting Ensemble Method** [\[paper\]](#)
 Sivasankar Elango, Selvi Chandran, *Mahalakshmi Sabanayagam*
 Journal of Soft Computing 2019
1. **Cross Domain Sentiment Analysis Using Different Machine Learning Techniques** [\[paper\]](#)
Mahalakshmi Sabanayagam, Sivasankar Elango
 Fifth International Conference on Fuzzy and Neuro Computing (*FANCCO 2015*)
 Poster at Grace Hopper Celebration India (*GHCI 2016*)

PREPRINTS / UNDER REVIEW

4. **Why Does Your Graph Neural Network Fail on Some Graphs? Insights from Exact Generalization Error.** Nil Ayday, *Mahalakshmi Sabanayagam*, Debarghya Ghoshdastidar.
3. **Generalization Certificates for Adversarially Robust Bayesian Linear Regression.** *Mahalakshmi Sabanayagam*, Russell Tsuchida, Cheng Soon Ong, Debarghya Ghoshdastidar.
2. **Cluster Specific Representation Learning.** *Mahalakshmi Sabanayagam*, Omar Al-Dabooni, Pascal Esser

*Authors contributed equally to the work

1. **Machine learning-based image detection for lensless microscopy in life science.** *Mahalakshmi Sabanayagam*, Jan Brunckhorst, Andreas Pirchner, Nikhitha Radhakrishna Naik [\[link\]](#)

RESEARCH ACTIVITIES

- *Workshop*: Understanding Generalization in Deep Learning, Burghausen, Germany February 19-21, 2025
- *Workshop*: AI in Science Conference, Canberra, Australia November 6, 2024
- *Summer School*: Statistical Physics & Machine Learning, Cargese, France August 1-12, 2023
- *Workshop*: Physics for Neural Networks, Princeton, USA April 17-19, 2023
- *Summer School*: Statistical Physics & Machine Learning, Les Houches, France July 4-29, 2022
- *Reviewer*: ICML 2026, 2025, ICLR 2025, NeurIPS 2024, TMLR 2024, AISTATS 2023, 2026

TEACHING / STUDENT JOBS

- *Teaching Assistant* for Seminar on Theoretical Advances in Deep Learning (WS 2022/23, WS 2023/24 WS 2024/25), Statistical Foundations of Learning (SS 2022), Analysis of new phenomena in machine/deep learning (SS 2022, SS 2023, SS 2024, SS 2025), Gems of Informatics 3: Modelling and analysis of graphs (WS 2021/22, WS 2022/23), Efficient Algorithms & Data Structures (WS 2020/21)
- *Student Supervision* **M.Sc. Thesis**: Vishnuraj Mavilodan, Anurag Singh, Admir Krnic, Ajinkya Mohgaonkar (ongoing); **M.Sc. Guided Research**: Alicia Dimroth, Ata Keskin (ongoing); **B.Sc. Thesis**: Michael Timothy
- *Research Assistant* in Certifiable AI at Fraunhofer-Institute, Munich (Sept 2020 – Feb 2021): Worked on novel ways to quantify risk in object detectors
- *Working Student* in Innovation Department at Osram Gmbh, Munich (Sept 2019 – Dec 2019): Developed faster RCNN and YOLO based models for detection, identification and tracking of multiple traffic objects

AWARDS & HONORS

- **Travel grant** for the First Workshop on Advances of Post-Bayesian Methods, London 2025
- **Best Paper Award** at the 3rd AdvML-Frontiers Workshop, NeurIPS 2024
- 3rd price in EMCR talk at AI in Science Conference 2024, Canberra, with a cash award of 1,000 AUD
- *Funding*: TUM-GS Internationalization Support of 3,000 Euro for the NYU research visit in 2023
- Largest sustainability impact award by **Siemens AI@sustainability Hackathon, 2020** for the AI solution towards finding new strategies that reduce the spread of COVID-19
- 2nd place in Female Tech Leaders Hackathon on **Introduction to Big Data: COVID-19 and its Global Effects, 2020** for analysing COVID-19 related tweets and the impact on equities
- Finalist in **Mobility Innovation Competition @ Campus, 2019** by Zentrum Digitalisierung Bayern
- **O.P. Jindal Engineering and Management Scholarship, 2012** one among 100 students all over India
- **Bachelor's Study scholarship** from NLC for the period 2011 – 2015
- Bronze medal (national level) and 1st in the city in **National Cyber Olympiad, 2007**

TECHNICAL SKILLS

Languages: Python (Proficient), C++ (Proficient), Java (Good)

Technologies: Tensorflow, Pytorch, JAX, NetworkX, Chromium Embedded Framework, OpenCV, AWS, Git